# 1. High Visceral Fat and Low SMI: Understanding Sarcopenic Obesity

## What does it mean?

Sarcopenic obesity is the coexistence of excess adiposity and low muscle mass or muscle function. The ESPEN/EASO consensus recommends assessing excess body fat alongside muscle function and body composition when evaluating this condition (Donini et al., 2022).

## How to interpret the InBody findings

| InBody finding | Interpretation |
|---|---|
| Visceral Fat Level ≥10 or Visceral Fat Area ≥100 $cm^2$ | Indicates elevated visceral adiposity relative to the applicable InBody reference values. |
| Low Skeletal Muscle Index (SMI) | May indicate low muscle mass when assessed using an appropriate calculation method and validated reference cut-off. |
| Both findings together | Suggests a body-composition pattern that warrants further evaluation for sarcopenic obesity. |

**Clinical assessment:** Muscle strength can be assessed using handgrip strength or other validated functional tests. The EWGSOP2 consensus identifies low muscle strength as the primary characteristic of sarcopenia, with low muscle quantity or quality used to confirm the diagnosis (Cruz-Jentoft et al., 2019).

## Understanding Skeletal Muscle Index (SMI)

For this document, SMI is calculated using appendicular skeletal muscle mass (ASM), which represents the combined muscle mass of the arms and legs.

**Formula:**

$SMI\ (kg/m^2) = ASM\ (kg) \div Height^2\ (m^2)$

The Asian Working Group for Sarcopenia (AWGS 2019) recommends the following BIA-based cut-offs for low appendicular skeletal muscle mass index in Asian adults:

| Sex | Low SMI |
|---|---|

| | |
|---|---|
| Men | Less than 7.0 kg/m² |
| Women | Less than 5.7 kg/m² |

**Application to InBody data:** Segmental Lean Analysis provides lean-mass values for the arms and legs. However, the sum of these segmental lean values should not automatically be assumed to represent validated ASM for clinical classification. A published 2025 letter has highlighted discrepancies between segmental lean totals and the reported skeletal muscle mass on InBody reports.

For research purposes, the muscle-mass variable, calculation method and reference cut-offs should therefore be specified and validated before classifying participants as having low SMI. This is particularly important when analysing pre-operative and post-operative bariatric patients.

# 2. Understanding Segmental Lean and Segmental Fat Analysis

## Segmental Lean Analysis

Segmental Lean Analysis provides information about lean mass across five body segments: right arm, left arm, trunk, right leg and left leg.

| Parameter | What it indicates |
|---|---|
| Mass (kg) | Lean mass measured for the respective body segment. |
| Percentage (%) | Comparison of the segment's lean mass with the device's reference value. |
| Under, Normal or Over | Indicates how the measured value compares with the applicable reference range. |

On applicable InBody models, approximately 90–110% represents the reference range for segmental lean mass. The exact range should be confirmed using the interpretation guide for the specific model.

## Segmental Fat Analysis

Segmental Fat Analysis displays fat mass for the five body segments.

| Parameter | What it indicates |
|---|---|

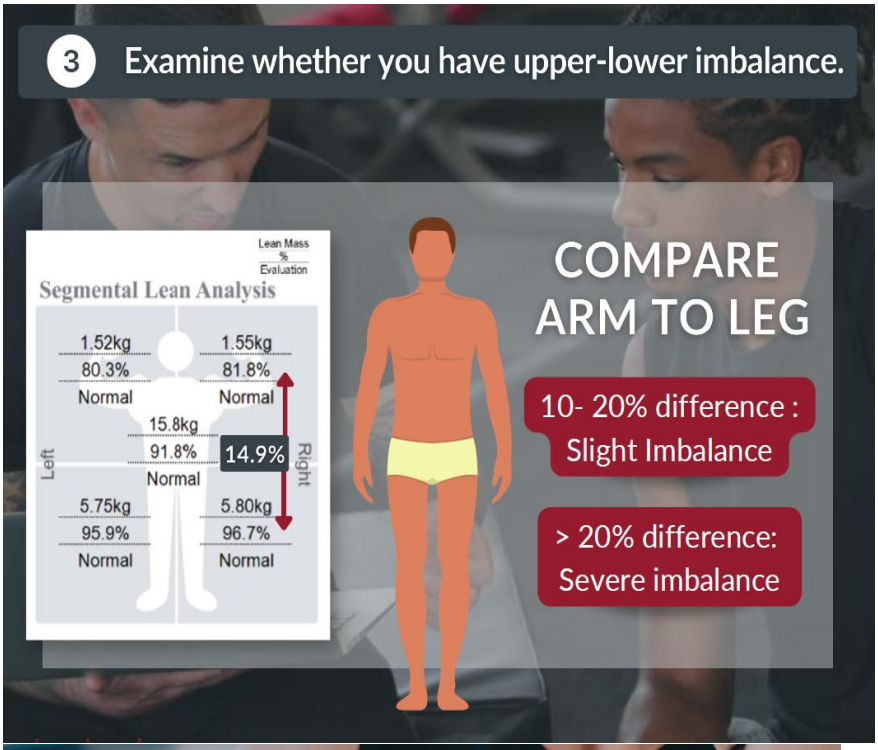
| | |
|---|---|
| Fat Mass (kg) | Estimated fat mass in the respective body segment. |
| Percentage (%) | Comparison with the device's reference value for segmental fat mass. |
| Under, Normal or Over | Indicates the measured value relative to the applicable reference range. |

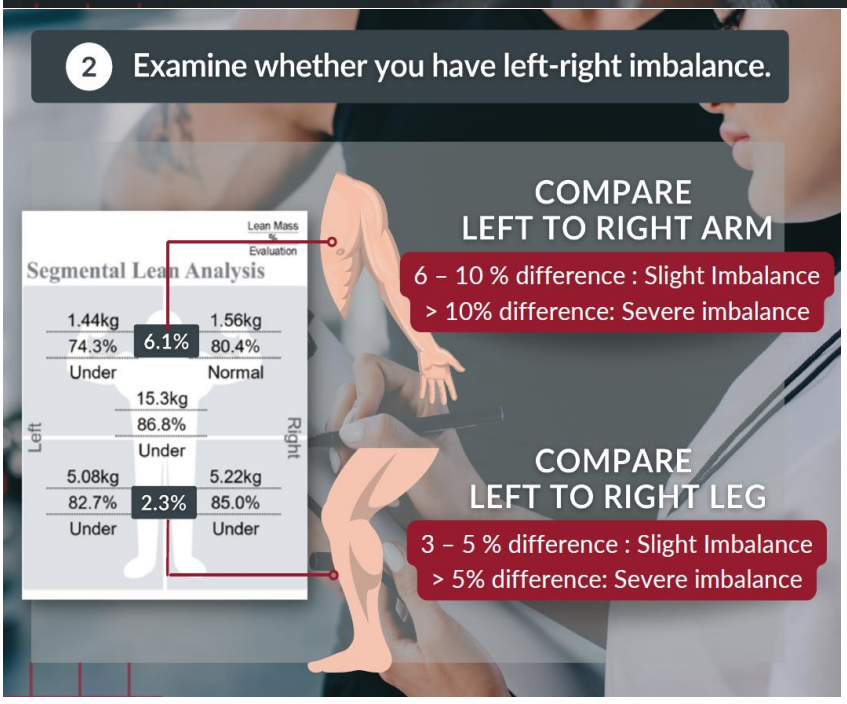
On applicable InBody models, approximately 80–160% represents the reference range for segmental fat mass.

BIA estimates body composition through electrical impedance and model-specific equations. Hydration and fluid distribution can influence estimates; therefore, consistent measurement conditions are important when comparing results over time (Kyle et al., 2004).

## Understanding Left–Right Imbalance

The displayed imbalance percentage represents the difference between the left and right segmental percentages, rather than the difference in mass in kilograms.




| Segment | Balanced | Slight imbalance | Severe imbalance |
|---|---|---|---|
| Arms | Below 6% | 6–10% | Above 10% |
| Legs | Below 3% | 3–5% | Above 5% |

# 3. Understanding the Percentages in Muscle-Fat Analysis

Muscle-Fat Analysis displays three parameters: Weight, Skeletal Muscle Mass (SMM) and Body Fat Mass (BFM). The percentages compare measured values with the device's standard values based on the entered height and sex.

| Parameter | Reference range | Interpretation |
|---|---|---|
| Weight | Approximately 85–115% | Compares measured weight with standard weight. |
| Skeletal Muscle Mass | Approximately 90–110% | Compares measured SMM with standard SMM. |
| Body Fat Mass | Approximately 80–160% | Compares measured fat mass with standard fat mass. |

These are general reference ranges for applicable report formats and should be checked against the relevant model's guide.

These percentages indicate the measured value relative to the device's reference or standard value. They are not population percentiles and do not indicate an individual's rank within a population.

## Understanding D-, I- and C-shaped Patterns

| Pattern | General appearance | Interpretation |
|---|---|---|
| D-shape | The muscle bar is relatively longer than the weight and fat bars. | Relatively higher muscle mass compared with the device's reference values. |
| I-shape | The three bars appear relatively similar. | A comparatively balanced pattern across the displayed parameters. |
| C-shape | The weight and fat bars are relatively longer than the muscle bar. | May indicate relatively higher fat mass with lower muscle mass compared with the reference values. |

These shapes are visual interpretation aids, not independent clinical diagnostic categories. The actual measurements and relevant clinical criteria should guide interpretation.

A C-shaped pattern accompanied by elevated visceral fat and low SMI may warrant further evaluation for sarcopenic obesity, consistent with the ESPEN/EASO recommendation to assess adiposity, muscle mass and muscle function together (Donini et al., 2022).

# 4. Understanding Target Weight, Fat Control and Muscle Control

## How is standard weight calculated?

For the applicable InBody report format, the following values illustrate a reference approach to standard weight and body composition.

| Parameter | Male | Female |
|---|---|---|
| Reference BMI | 22 kg/m² | 21.5 kg/m² |
| Standard weight | 22 × height² (m²) | 21.5 × height² (m²) |

## What do Fat Control and Muscle Control indicate?

- **Fat Control:** Indicates the suggested change in fat mass towards the device's target body composition.
- **Muscle Control:** Indicates the suggested change in muscle mass towards the device's target body composition.

These values support body-composition assessment beyond body weight alone. The exact internal algorithm used to generate these recommendations is not established by the research references below; the values should therefore be interpreted according to the applicable device guidance.

# References


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